

Step	Step Description	Agents Involved	Tool Calls and Parameters	Reason
1	User asks: Which buildings in Exeter already have solar panels on roofs?	human, host_agent	Message from human to host_agent	User initiates the query.
2	Plan steps to solve the query	planning_agent	send_message to planning_agent with task_description	planning_agent is chosen to break down the geospatial query into actionable steps, as per its prompt.
3	Planning agent outputs steps: ["Find Exeter 1 area", "Search for buildings in Exeter with solar panels on roofs as many search results"]	planning_agent	Output steps	Steps are based on the query: first, find the area polygon for Exeter, then search for buildings with solar panels.
4	Find the area polygon for Exeter	named_area_agent	send_message to named_area with task_description	named_area_agent is chosen because the prompt specifies it provides polygons for named areas like Exeter.
5	Retrieve named area polygons (including Exeter)	named_area_agent	call_os_ngd with polygon_or_point=True, filename='exeter_area_polygon'	The named_area_agent uses the OS NGD tool to get all named area polygons, as per its API capabilities.
6	Filter to the specific Exeter polygon	data_analysis_agent	send_message to data_analysis_agent with task_description to filter 'exeter_area_polygon' for Exeter	Filtering is needed because the initial artifact contains multiple named areas; only the one for Exeter is relevant.
7	Generate metadata for 'exeter_area_polygon'	data_analysis_agent	generate_metadata_for_artifacts with artifact_names=['exeter_area_polygon']	Metadata is generated to understand which columns contain the area name, as per the coding agent's guidelines.
8	Code executor filters polygons for Exeter	code_executor	Function filters polygons where any name column contains 'Exeter' (case-insensitive)	The code uses all name columns (name1_text, name2_text, etc.) based on metadata, ensuring robust filtering for 'Exeter'.
9	Save filtered Exeter polygon	data_analysis_agent	Output: filtered_exeter_area_polygon	The result is a GeoDataFrame with only the Exeter polygon, as required for further spatial queries.
10	Search for buildings in Exeter with solar panels on roofs	buildings_agent	send_message to buildings with task_description to search in 'filtered_exeter_area_polygon'	The buildings_agent is chosen because the prompt specifies it searches for buildings within a given area.
11	Query buildings database for buildings in Exeter	buildings_agent	call_os_ngd with bbox='filtered_exeter_area_polygon', filename='exeter_buildings_with_solar_panels'	The bbox parameter is set to the filtered Exeter polygon to restrict the search to Exeter.
12	Filter buildings for solar panels on roofs	data_analysis_agent	send_message to data_analysis_agent to filter artifacts for solar panels on roofs	Further filtering is needed because the initial search may not directly filter for solar panels; the coding agent is used for this.
13	Generate metadata for building artifacts	data_analysis_agent	generate_metadata_for_artifacts for relevant building artifacts	Metadata is needed to identify the correct column for solar panel presence.
14	Code executor filters buildings/building parts with solar panels	code_executor	Function filters for 'roofmaterial_solarpanelpresence' == 'Present' in buildings and building parts	The code uses the metadata to select the correct column and combines results from both buildings and building parts.
15	Save filtered buildings with solar panels	data_analysis_agent	Output: filtered_solar_panel_buildings_exeter	The result is a GeoDataFrame of buildings/building parts in Exeter with solar panels.
16	Search for solar panel structures in Exeter	structures_agent	send_message to structures_agent to search in 'filtered_exeter_area_polygon'	The structures_agent is chosen because the prompt specifies it searches for non-building structures, such as solar panel installations.
17	Query structures database for solar panel structures	structures_agent	call_os_ngd with filters for solar panel-related structures, bbox='filtered_exeter_area_polygon', filename='solar_panel_structures_exeter'	Filters are set to include all relevant solar panel structure types, as per the agent's prompt and available filters.
18	Analyze and summarize solar panel structures	data_analysis_agent	send_message to data_analysis_agent to filter and summarize 'solar_panel_structures_exeter'	The coding agent is used to further filter and summarize the structures, as the initial search may include spurious entries.
19	Generate metadata for solar panel structures artifact	data_analysis_agent	generate_metadata_for_artifacts for 'solar_panel_structures_exeter'	Metadata is needed to identify columns for filtering and summarization.
20	Code executor filters and summarizes solar panel structures	code_executor	Function filters using multiple columns (description, oslandcovertierb, oslandusetierb, nlud_groupdescription) for solar panel relevance	Multiple columns are used for robust filtering, as per the coding agent's guidelines and metadata.
21	Save filtered solar panel structures	data_analysis_agent	Output: filtered_solar_panel_structures_exeter	The result is a GeoDataFrame with summary statistics for solar panel structures in Exeter.
22	Plot buildings and structures with solar panels on Exeter map	plotting_agent	send_message to plotting_agent to plot 'filtered_solar_panel_buildings_exeter', 'filtered_solar_panel_structures_exeter', 'filtered_exeter_area_polygon'	The plotting_agent is chosen to visualize the results on a map, as per its prompt.
23	Generate metadata for plotting artifacts	plotting_agent	generate_metadata_for_artifacts for plotting artifacts	Metadata is needed to ensure correct columns and CRS for plotting.
24	Code executor generates folium map	code_executor	Function creates a folium map with Exeter boundary, buildings/building parts with solar panels, and solar panel structures	The code uses the correct CRS and plots all relevant features, as per the plotting agent's guidelines.
25	Output summary and map artifact	plotting_agent	Output: summary, map filename, description, map file	The result is a summary and an interactive map showing all relevant features.
26	Host agent presents final results to user	host_agent	Output: summary of findings and map availability	The host_agent summarizes the results for the user, as per its role.

Overall Summary: - Databases used: Named area (for Exeter polygon), buildings (for buildings and building parts), structures (for solar panel structures). - Filters/code: Named area polygons were filtered using all name columns for 'Exeter'. Buildings and building parts were filtered for 'roofmaterial_solarpanelpresence' == 'Present'. Structures were filtered using multiple columns for solar panel relevance. All results were spatially restricted to the Exeter polygon. - Final outcome: 4,468 buildings/building parts in Exeter have solar panels on roofs; 1 large solar panel structure ("Solar Panels On Bare Earth Or Grass") was found. An interactive folium map was generated showing all results. - Overall reasoning: Each agent and tool was chosen based on its prompt and capabilities. Metadata was used to select correct columns for filtering. Code was written to robustly filter and summarize results, and plotting was done using folium with correct CRS and artifact handling. The process ensured accurate, spatially-restricted, and user-friendly results.